

Measures of association & impact

A quick visual reference for quantifying exposure-outcome relationships

Measures of association

Compare the risk or odds of an outcome between an exposed and an unexposed group.

Measures of impact

Translate that association into real-world consequences: how many cases, how much benefit.

Four core measures

Relative risk (RR)

Risk in exposed divided by risk in unexposed

Odds ratio (OR)

Odds in exposed divided by odds in unexposed

Attributable risk (AR)

Risk difference: how much exposure adds

Number needed to treat

Patients treated for one extra good outcome

The 2 x 2 table behind every measure

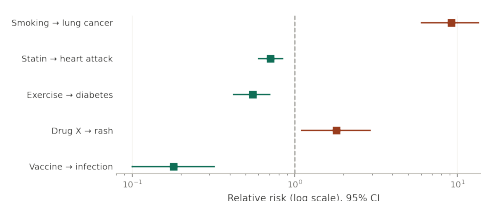
	Outcome present	Outcome absent
Exposed	a exposed with outcome used in numerator of RR and OR	b exposed without outcome completes exposed group
Unexposed	c unexposed with outcome used in denominator of RR	d unexposed without outcome completes unexposed group

From the table above

$$RR = [a / (a+b)] / [c / (c+d)] \quad OR = (a \times d) / (b \times c)$$

$$AR = \text{risk}(\text{exposed}) - \text{risk}(\text{unexposed}) \quad NNT = 1 / AR$$

Forest plot: comparing several relative risks at a glance



Quick reference: RR/OR above 1 means harm, below 1 means protection; the dashed line at 1 is no effect.